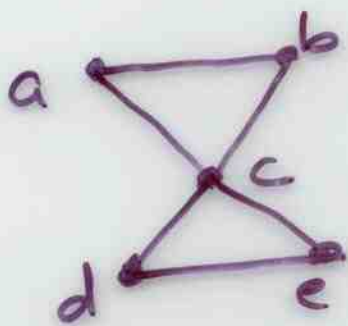


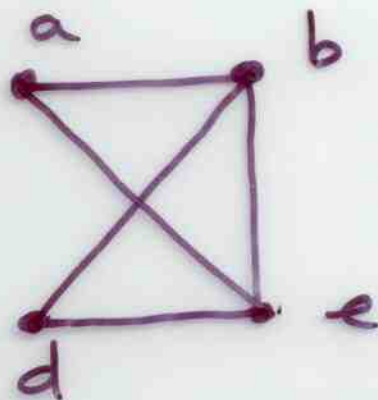
Euler Cycles

Defn: An Euler Cycle (or Circuit) is a cycle that traverses every edge of a graph exactly once.

If there is an open path that traverses each edge only once, it's called Euler's path.

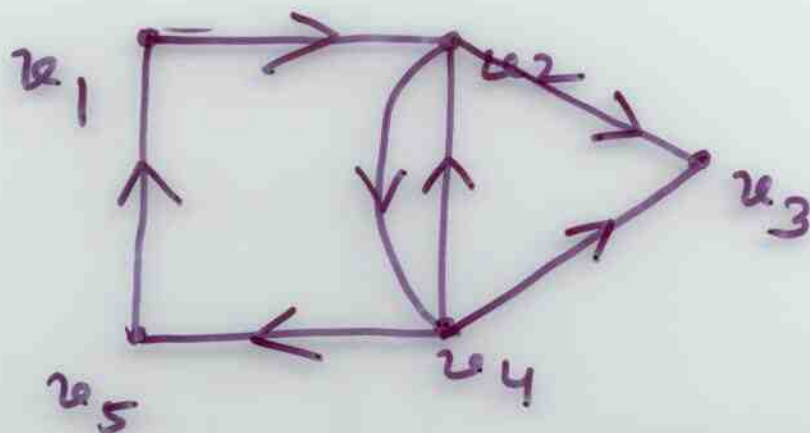


a c d e c b a
Euler cycle.



b a e d b e
Euler path

In a directed graph it will be less likely to have an Euler path or circuit because you must travel in the correct direction.



Neither Euler circuit
nor Euler path

Necessary and sufficient condition for an Euler ckt -

Thm! A connected, undirected multigraph has an Euler ckt iff each of its vertices has even degree.

Proof! ~~in~~ in ~~the~~ ckt every edge is being traversed exactly once. While traveling the ckt we move into a vertex by one edge and leave by another edge. So there must be an even No. of edges adjacent to each vertex.

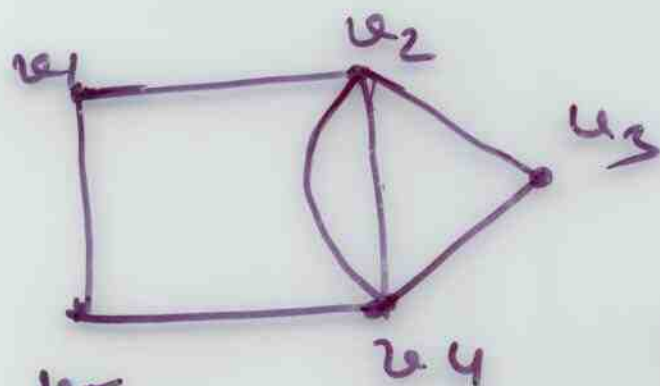
(2)

Thm!:- A connected, undirected multigraph has an Euler path but not an Euler ckt iff it has exactly two vertices of odd degree.

Hamilton ckt's:

Def! A Hamilton path is a path in a graph G that passes through every vertex exactly once.

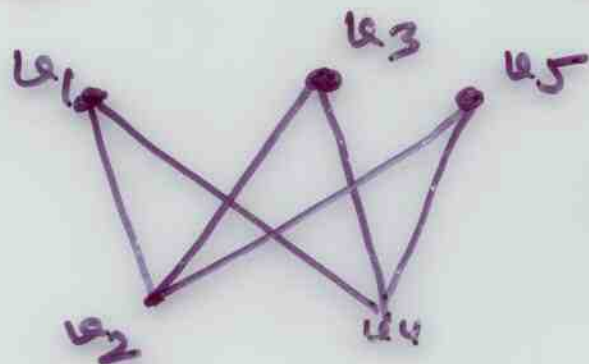
Def! A Hamilton ckt is a Hamilton path that is also a ckt.



Hamilton ckt
(path to)

$v_1 v_2 v_3 v_4 v_5 v_1$

Ex



No Hamilton ckt
but
 $v_1 v_2 v_3 v_4 v_5$ is
a Hamilton path

(3)

Sufficient Condition for a Hamilton Ckt

Let G be connected, simple graph with N vertices, where $N \geq 3$

If the degree of each vertex is at least $n/2$ then G has a Hamilton circuit.

Note: Unfortunately there are no necessary and sufficient conditions to determine if a graph has a Hamilton ckt and/or path. Fortunately there are theorems that give sufficient conditions for the existence of a Hamilton ckt.